

Purvi Verma

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Summary

AI Engineer with 2+ years of hands-on experience shipping production LLM and multimodal AI systems, from safety guardrails and agentic workflows to computer-vision pipelines processing thousands of enterprise assets. Skilled across the GenAI stack (LangChain, LangGraph, RAG, vLLM) with strong MLOps/LLMOps practices for monitoring, versioning, and retraining models in production. Currently pursuing an M.Tech in AI at IIT Roorkee to deepen research and technical foundations.

Technical Skills

Languages: Python, C++, Julia, SQL

ML/AI: LLMs, NLP, Computer Vision, Multimodal AI, Model Fine-Tuning (LoRA, PEFT), Prompt Engineering

ML Platforms: Vertex AI (Gemini), Azure AI, MLflow – experiment tracking, model registry, A/B testing

LLM Stack: LangChain, LangGraph, RAG, MCP, Vector Databases (FAISS/Pinecone/Chroma)

Frameworks: PyTorch, TensorFlow, Hugging Face, DeepSpeed (distributed training)

MLOps / LLMOps: Docker, Kubernetes, CI/CD pipelines, model monitoring, retraining workflows, model evaluation & benchmarking

Professional Experience

Senior Associate, Data Science L1

Bangalore, India

Publicis Sapient

July 2024 – July 2026

- Identified critical AI safety gaps risking non-compliant outputs in enterprise LLM deployments; architected AI safety guardrails for Bodhi Compliance (Bodhi platform), cutting harmful output rates from 45% to under 5%.
- Recognized limitations in single-turn LLM interactions for complex enterprise workflows; designed a multi-turn LLM agent with tool calling and context-aware orchestration, raising task success rate from 70% to 92% and enabling systematic A/B testing across model variants.
- Addressed the lack of automated, accurate document processing for enterprise digital assets; engineered Bodhi Vision, an end-to-end multimodal (CV + LLM) pipeline with intelligent image format conversion (PNG/JPEG/WebP to PSD), vLLM-powered segmentation, and advanced text detection replacing a fully manual process with one achieving 90.3% extraction accuracy, 87% semantic consistency, and zero failures across 500+ enterprise assets.
- Tasked with scaling multimodal AI for regulated pharmaceutical marketing content; deployed a production CV + LLM pipeline for Eli Lilly, moving asset processing from manual review to automated handling of 10,000+ pharmaceutical assets with high accuracy in a compliance-sensitive environment.

Artificial Intelligence Intern

Remote

Blockatena

March 2023 – May 2023

- Identified that single-image future-frame prediction lacked reliable temporal modeling, causing generated sequences to lose coherence over successive frames; researched and implemented a hybrid GAN + LSTM architecture to jointly learn spatial features and temporal dependencies.
- Trained the generator-discriminator setup on image sequence data to forecast multiple future frames from a single input image, iterating on network architecture and hyperparameters to improve prediction quality; improved temporal consistency in generated sequences compared to a baseline GAN without sequence modeling.

Projects

CodeMap – AI-Powered Code Architecture Analyzer

- Built a 7-stage pipeline combining static analysis with LLM reasoning to auto-generate architecture diagrams, dependency graphs, and docs from any repo, cutting legacy codebase onboarding time.
- Engineered batched LLM summarization (10 functions/call) and automated design-pattern/anti-pattern detection (god classes, facades, layered architectures) for high-accuracy code summaries.
- Implemented self-healing Mermaid diagram generation where an LLM auto-repairs invalid syntax before rendering, completing full end-to-end analysis in under 5 minutes.

Safeguard Toolkit – Open-Source Python Security Scanner

`pip install safeguard_toolkit`

- Designed a modular **ScanManager** architecture with 4 composable scanners detecting hardcoded secrets, risky configs, vulnerable dependencies, and unsafe file permissions in Python codebases.
- Built entropy-based secret detection and a dependency checker parsing `requirements.txt/Pipfile/pyproject.toml` to flag outdated and license-incompatible packages.
- Packaged and published to PyPI with CI/CD-ready plugin support, enabling teams to catch vulnerabilities pre-deployment across the software supply chain.

Education

M.Tech in Artificial Intelligence	2026 – 2028 (In Progress)
Indian Institute of Technology (IIT), Roorkee	
B.Tech	2020 – 2024
Indian Institute of Technology (IIT), Bhubaneswar	CGPA: 8.2/10
Class XII, CBSE Board	2020
	Percentage: 92.2%
Class X, CBSE Board	2018
	Percentage: 91.6%

Open Source Contributions

- **Google Bespoke:** Improved error handling and API key resolution in an LLM/TTS client library, replacing opaque HTTP exceptions with descriptive, debuggable error messages.
- **video-keyframe-detector:** Replaced print-based output with structured logging and fixed core algorithm bugs (uninitialized frame handling, frame-count safeguards, CSV output integrity) in a computer vision keyframe extraction tool.
- **CRAFT-pytorch:** Updated OCR text-detection model to support current PyTorch/torchvision versions and added automatic CUDA/CPU device fallback for broader hardware compatibility (*PR pending review*).

Scholastic Achievements

- Google Cloud Agentic AI Day Hackathon – Finalist, Top 700 out of 9,100 teams (2025); built AI-Powered Agricultural Assistant, a multimodal AI assistant for Indian farmers.
- Real-time Age & Gender Detection System – 1st Place, Hackathon at IIT Bhubaneswar (2023).